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| **Fiscal Year:** | FY 2024 | **Task Last Updated:** | FY 10/04/2023 |
| **PI Name:** | Settles, Andrew Mark Ph.D. | | |
| **Project Title:** | Understanding Genome-Wide Mutation Load in Spaceflight Culture of Cyanobacteria | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Microbiology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | (1) Bioregenerative Life Support | | |
| **PI Email:** | andrew.m.settles@nasa.gov | **Fax:** | FY |
| **PI Organization Type:** | NASA CENTER | **Phone:** | 352-283-2767 |
| **Organization Name:** | NASA Ames Research Center | | |
| **PI Address 1:** | AST Life Support Studies | | |
| **PI Address 2:** | NASA Ames Research Center | | |
| **PI Web Page:** | | | |
| **City:** | Moffett Field | **State:** | CA |
| **Zip Code:** | 94035-1000 | **Congressional District:** | 18 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research |
| **Start Date:** | 10/01/2021 | **End Date:** | 09/30/2024 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Griko, Yuri | **Contact Phone:** | 650-604-0519 |
| **Contact Email:** | Yuri.V.Griko@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 09/30/2024 per F. Hernandez/ARC (Ed., 10/10/23).<br>NOTE: Project dates for this investigation have changed; the period of performance (POP) is now 10/1/2021-09/30/2023 per F. Hernandez/ARC. Original POP was 09/1/2021 - 08/31/2024 (Ed. 1/19/22). | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | Internal Project | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Increased radiation and the microgravity environment of spaceflight is expected to increase genetic mutation of all organisms. Biological Life Support systems incorporate organisms that can fix carbon dioxide and nitrogenous wastes into sugars, proteins, and other carbohydrates to release oxygen and provide feedstocks for food, fuel, or manufacturing. Plants and microbes can be used for biological support, and it is essential to understand how spaceflight impacts mutation rates in these organisms to enable appropriate countermeasures to ensure high life support productivity during long duration and deep space missions. This project will focus on the cyanobacterium Arthrospira platensis. Commonly known as Spirulina, this is an edible microalgae that is a candidate for biological life support. The project will grow A. platensis in serial cultures to allow the organism to evolve in spaceflight on the International Space Station. Whole genome sequencing of culture samples will determine the mutation load of the cyanobacteria at multiple time points throughout the long-term growth experiment. Multi-omics analysis including RNA sequencing, proteomics, metabolomics, and analytical chemistry of nutritional composition will give insights on the biological responses of the cells to spaceflight. The project directly addresses research topic 1 (Microbiology Studies in Support of Space Exploration ) by studying a microorganism that is a candidate to support long-duration space exploration. The experiments proposed will reveal whether long-term growth of A. platensis will be simple to implement or if there is risk of undesirable cellular responses that require mitigation. If spaceflight is a significant stress for this species, the project will identify specific genes and pathways that would need to be targeted for genetic improvement. The project is expected to further our knowledge of microorganism adaptation to long-term growth in spaceflight and contribute towards the development of bioregenerative life support systems. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This project primarily investigates DNA damage risks associated with low Earth orbit. The focus is on an algae crop that is grown as a protein source in human and animal diets. The fundamental knowledge developed in the project will contribute to a better understanding of DNA damage mechanisms and may reveal new targets for protecting humans or agriculture from environmental stresses. |
| **Task Progress:** | We completed flight definition experiments to establish our science requirements and basic concept of operations for the flight experiment. The flight definition experiments established the following science requirements:<br>1) Culture growth parameters: We determined growth parameters for the NIES-39 strain of A. platensis to result in a 13-to-15-day growth cycle. This growth cycle is slower than optimal growth in order to reduce experiment mass and volume and to limit crew time to eleven serial passages for 5-to-6-month exposure to spaceflight. We optimized the media composition, light intensity, growth temperature, inoculum quantity, and bioreactor bag material.<br><br>2) Live cell storage conditions: We developed a method for storing A. platensis inoculated cultures alive in bioreactor bags in dark, soft stowage conditions for up to 10 weeks prior to transfer to lighted growth conditions. This method allows the spaceflight experiment to be initiated at NASA Ames Research Center and integrated into a resupply flight without a late load requirement. Initial experiments compared additions of different carbon sources to the media, dry storage of cell pellets, and a variety of temperature conditions.<br><br>3) Media shelf life: We determined that Zarrouk's media has at least a 6-month shelf life when stored in flurorinated ethylene propylene (FEP) bioreactor bags either as liquid media or in dry salt form.<br><br>4) Cell morphology imaging: We confirmed that A. platensis trichome structure can be imaged directly from cultures in FEP bioreactor bags.<br><br>5) Biomass harvest methods: We developed a filtration method to dewater cultures and concentrate biomass for storage in ultralow freezers. This method reduces the volume of frozen material that will need to be returned and simplifies biomass sample processing in the laboratory.<br><br>6) Cryopreservation of live cells: We tested a variety of methods to cryopreserve A. platensis cultures and confirmed that adding 10% dimethyl sulfoxide (DMSO) to cultures according to Shiraishi (2016) resulted in viable frozen cells for short duration storage at -80°C. However, long-term viability of the cryopreserved cells declined more rapidly than reported. We are currently testing a variety of alternative cryoprotectants to screen for more robust protocols to return live cells from each passage of the experiment.<br><br>7) Multi-omics extraction methods: We tested a variety of sample processing methods for DNA, RNA, and protein extractions. Input biomass processes tested included frozen cultures, frozen dewatered biomass, and lyophilized biomass. Cell disruption methods included heat lysis and bead beating for different time intervals. In addition, multiple extraction protocols were used for DNA and RNA. DNA and RNA extractions were characterized for total quantity extracted per unit dry biomass, purity based on UV/Qubit spectroscopy, and molecular integrity using a bioanalyzer.<br><br>8) Metabolomics & Nutritional Analysis: We engaged a metabolomics laboratory service center to characterize untargeted metabolomics sample analysis for A. platensis. Our initial experiment will compare flask-grown cultures to FEP bioreactor bag cultures. Samples were submitted in July 2023, and analyzed data is expected in October 2023. We also completed a literature survey of spirulina nutritional composition to rank nutrients for priority of analysis based on human recommended daily values. |
| **Bibliography Type:** | Description: (Last Updated: 05/04/2026) |
| **Abstracts for Journals and Proceedings** | Settles AM, Fisher KE, Zhang J, Bai F, Beisel NS, Hersh H, Tyre KN, Müller BSF. "Space Algae: Understanding the genomic impacts on microalgae after growth in the International Space Station." Plant & Animal Genome Conference 2023 (PAG 30), San Diego, CA, January 13-18, 2023.<br>Abstracts. Plant & Animal Genome Conference 2023 (PAG 30), San Diego, CA, January 13-18, 2023. Invited talk and poster presentation. , Jan-2023 |